ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M027 **Name:** Aishwarya Hegde

Questions attempted: Q1, Q2, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16\times10^{-3} / 5\times10^{-6}$ = **3200 px**.

(ii) Sensor width = 1280 × 5 μm = 6.4 mm. Using similar triangles, FOV = $W_s \cdot f / Z = 6.4 \times 16/400$ = **0.256 mm**.

Distance per pixel = FOV/1280 = 0.256/1280 = **2×10^{-4} mm** = 0.2 μm.

(b) LSB = 5/4096 = 1.2207 mV.

ΔF = LSB / S = 1.2207 mV / 10 mV/N = **0.0122 N**.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT} x(kT) + \int_0^T e^{A(T-s)} B\, u(kT)\, ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma} B\, d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity chek: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain G/(1 − F) = 0.090635/0.18127 = 0.5 ✓.

$$\mathbf{x_{k+1} = 0.8187\, x_k + 0.0906\, u_k}$$

Question 2

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\, x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so **$|\tau_0| \le mg\ell = 2.4525$ N·m**. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5$ s^{-1}.

A = [0, 1; −61.31, −2.5], and sence $\partial f_2/\partial\tau = 1/J = 50$, **B = [0; 50]**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to

$mg\ell \sin \theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torque would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torque would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a steady error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin \theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\ e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

$e = 0.04899$ rad $\times (180/\pi) = 2.807°$, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small becuase K_p is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

(d) With the compensation term, at rest $K_p(\theta_d - \theta_{ss}) + mg\ell \sin \theta_{ss} = mg\ell \sin \theta_{ss}$, so $\theta_{ss} = \theta_d$ and the steady-state error is **0**.

Note: K_d does not influence the steady-state angle because $\dot{\theta} = 0$ there.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\,r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain $= bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 =$ **0.8**.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 =$ **20%**. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Substituting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow$ **$k_2 = 2$**; $2 + k_1 = 24 \Rightarrow$ **$k_1 = 22$**; $K_I =$ **32**.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. **$\theta_{ss} = r$**.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrum (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the actuator is saturated, conditional integration, or back-calculation that feeds the diffrence between commanded and saturated torque back to the integrator.

Note: the integrator adds one state, so the loop becomes third order.